

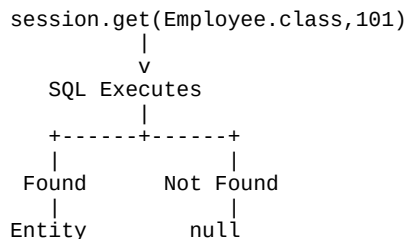
Chapter 5 - get(), load() and getReference()

1. Introduction

Hibernate provides different APIs to retrieve an entity. The commonly discussed methods are `Session.get()`, `Session.load()` (deprecated in Hibernate 6), and `EntityManager.getReference()`.

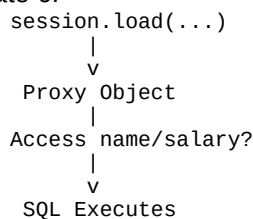
2. Session.get()

`get()` immediately executes a SELECT statement and returns the entity. If no record exists, it returns null.



3. Session.load()

`load()` returns a proxy object first. SQL is executed only when a non-ID property is accessed. If the record does not exist, `ObjectNotFoundException` is raised when the proxy initializes. Deprecated in Hibernate 6.



4. EntityManager.getReference()

`getReference()` is the JPA equivalent of lazy proxy retrieval. It returns a proxy first. When the proxy is initialized, Hibernate executes SQL. If the row is missing, `EntityNotFoundException` is thrown.

5. Proxy Object

A proxy is a lightweight placeholder for an entity. It initially contains only identifier information and loads the remaining state when required.

6. Comparison

Feature	get()	load()	getReference()
Framework	Hibernate	Hibernate	JPA
Initial SQL	Immediate	Lazy	Lazy

Initial Return	Entity	Proxy	Proxy
Not Found	null	ObjectNotFoundException*	EntityNotFoundException*
Hibernate 6	Supported	Deprecated	Supported

7. Code Example

```
Employee e1 = session.get(Employee.class,1);

Employee e2 = session.load(Employee.class,1);

Employee e3 = entityManager.getReference(Employee.class,1);
```

8. Interview Tips

Use `get()` when you need the entity immediately. Use `getReference()` when a proxy is sufficient, such as setting a relationship by ID without reading all columns. Remember that `load()` is deprecated in Hibernate 6.

9. Common Mistakes

Do not describe `get()` as a `FetchType.EAGER` feature. Eager/Lazy are primarily relationship fetch strategies. Also remember that if `get()` returns null, Hibernate does not cache null in the Level 1 Cache.

10. Interview Questions

1. Difference between `get()` and `load()`? 2. What is a proxy object? 3. Why is `load()` deprecated? 4. When is `getReference()` useful? 5. Which method returns null if a record is missing?